cells we take a rectangle obtained by stretching the cell further along all directions, and each of the sides will measure like the corresponding cell's side plus 2ε. This way, core points close to the cells' borders will have their ε-neighborhood contained in either shard.

This solution can therefore work limitedly to certain specific cases, in specific configurations, and hardly in high-dimensional datasets.

Nevertheless, the basic idea is valid, and we can make it work by simply changing the way we shard the dataset: instead of relying on random sharding or spherical canopies, we can break the dataset into a regular grid, where each cell is of the same size. Cells are hyper-rectangles instead of hyper-spheres, and for each coordinate the domain can be split differently, causing the cells (rectangles)' sides to be each of a different length.

This is a good improvement over dealing with canopies, because identifying points close to the borders becomes easy: we can just check that the absolute value of the difference between the point and the border along some coordinate is smaller than a threshold, instead of computing the distance between a point and a hyper-sphere; moreover, it becomes easier and cheaper to shard the dataset into grid cells.

But… it gets even better! Instead of having to compare points close to the borders of adjacent cells, we can define the cells to be slightly overlapping, and precisely to be overlapping, for each coordinate, over a rectangular area of side ε: as shown in figure 13.12, this way each adjacent cell